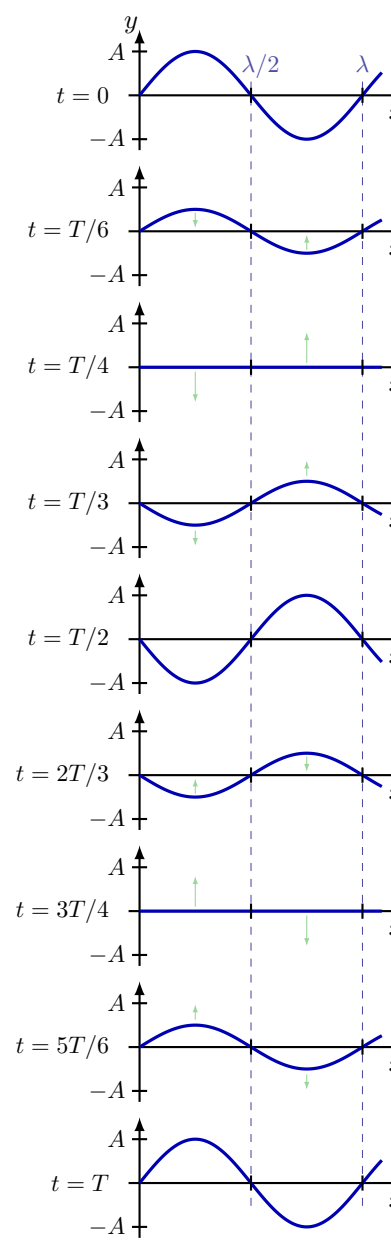
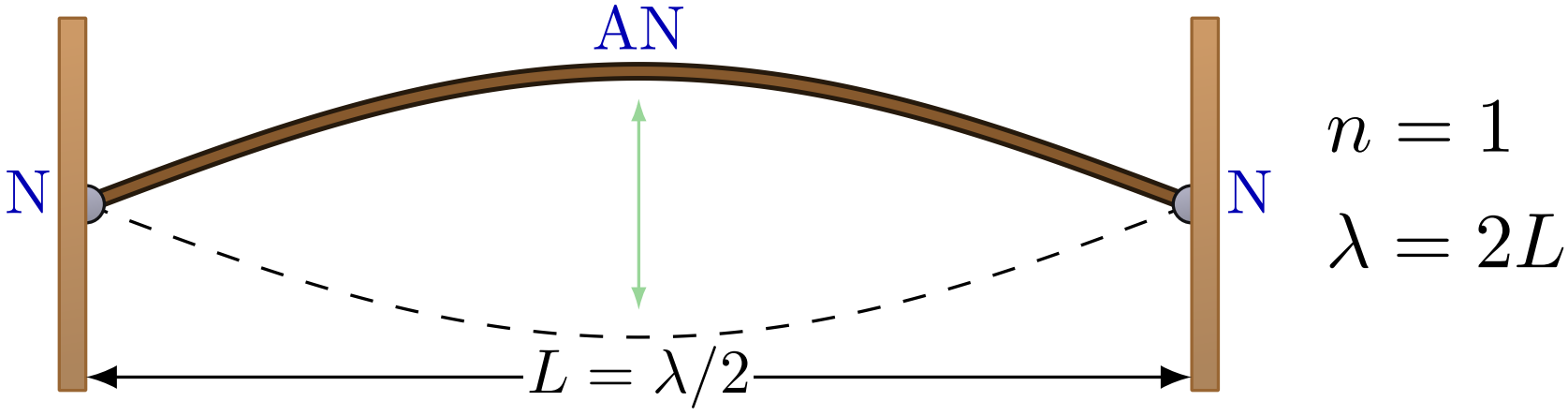
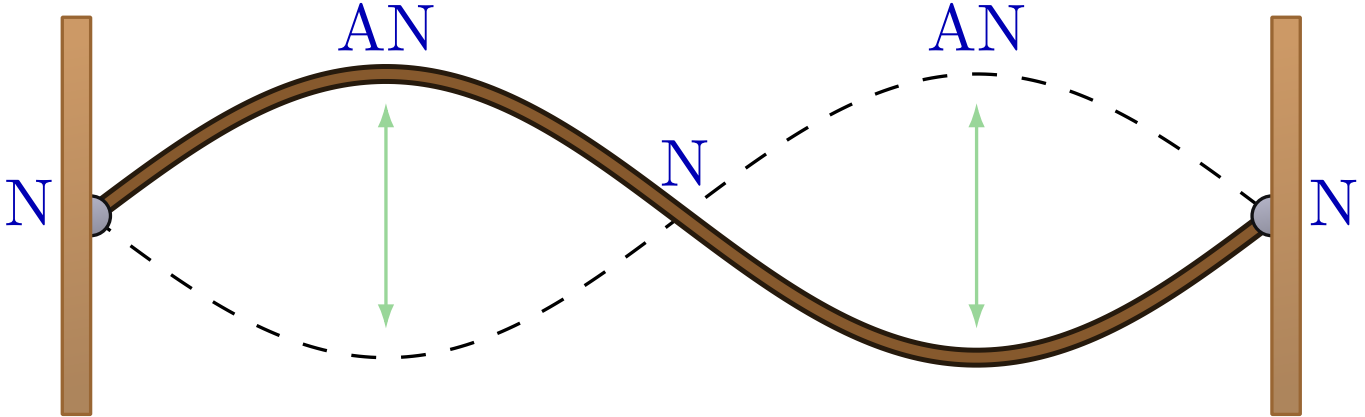

y
$y(t) = A\sin(kx - \omega t)$
A
$-A$
v
x
+
y
$y(t) = A\sin(kx + \omega t)$
A
$-A$
v
x
=
y
$y(t) = 2A\sin(kx)\sin(\omega t)$
$2A$
$-2A$
x
$v_x = 0$

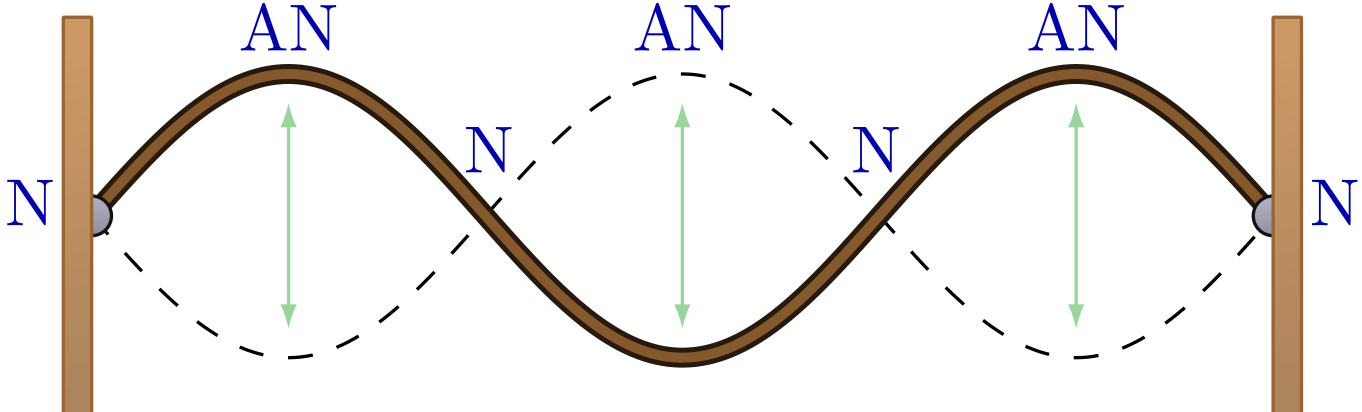

y
A
−A
λ/2
λ
x
t = 0
t = T/6
t = T/4
t = T/3
t = T/2
t = 2T/3
t = 3T/4
t = 5T/6
t = T


AN
N
N
$L = \lambda/2$
$n = 1$
$\lambda = 2L$



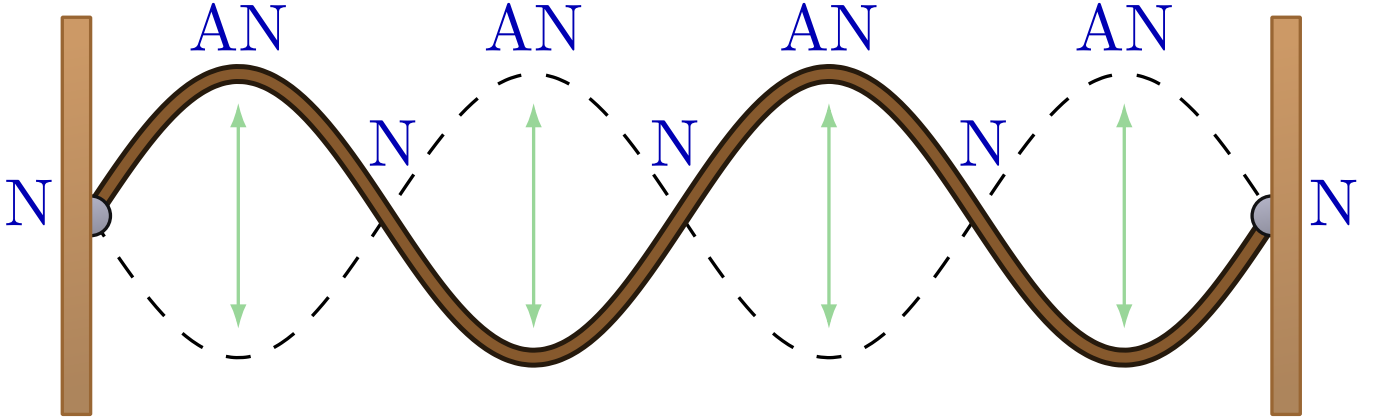
$n = 2$

$\lambda = L$



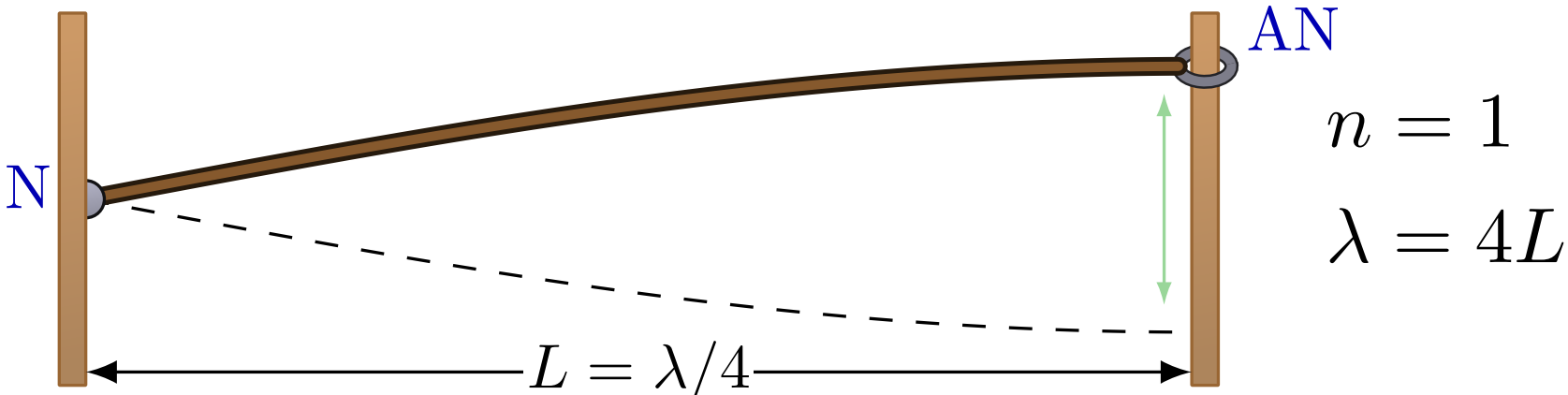
$$n = 3$$

$$\lambda = \frac{2L}{3}$$



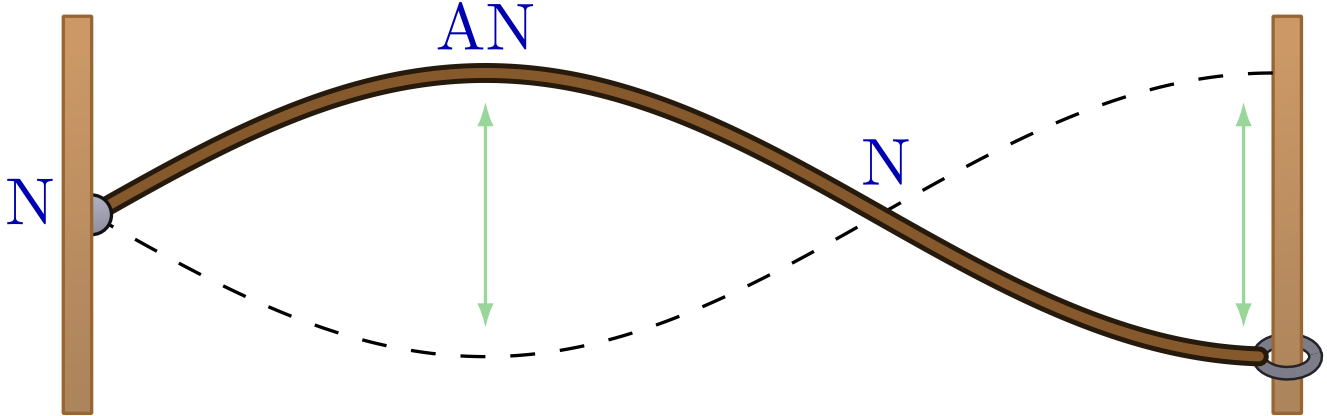
$$n = 4$$

$$\lambda = \frac{L}{2}$$



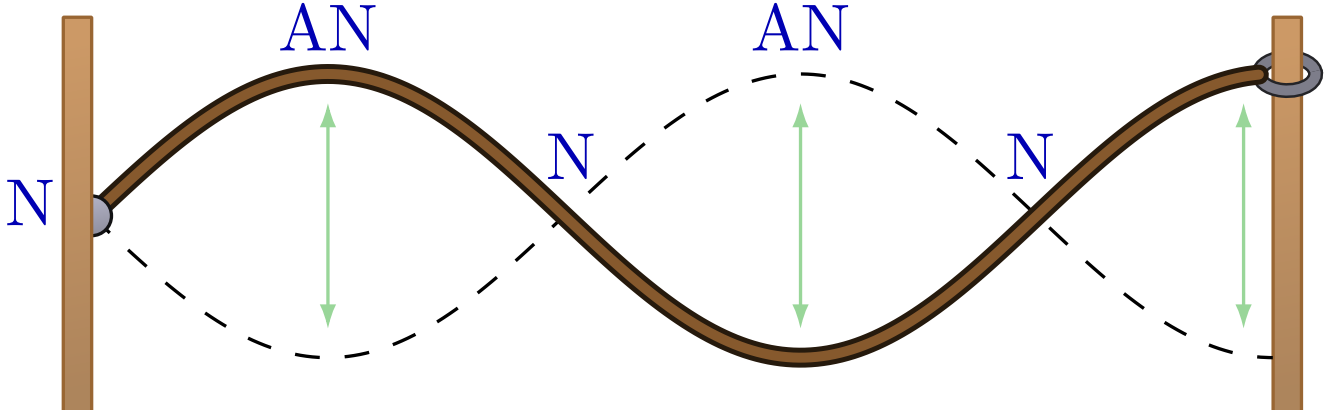
$$n = 1$$

$$\lambda = 4L$$



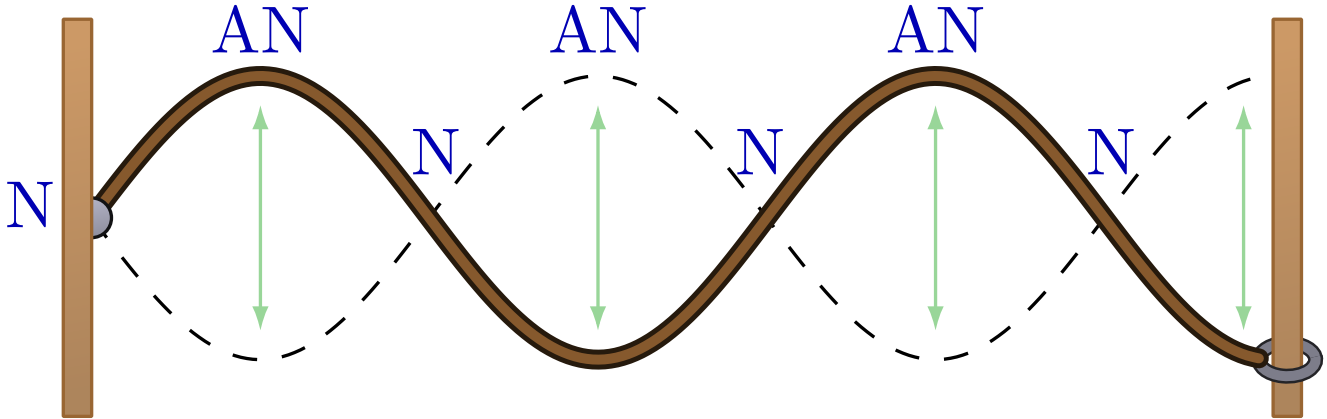
$$n = 3$$

$$\lambda = \frac{4L}{3}$$



$$n = 5$$

$$\lambda = \frac{4L}{5}$$



$$n = 7$$

$$\lambda = \frac{4L}{7}$$